

Solutions — IMC Training Sheet, May 27, 2026

Problem 1. (IMC 2008, Day 2, Problem 2) Two ellipses with a common focus have at most two common points.

Solution. An ellipse can be described as the locus of points X such that

$$\frac{|FX|}{\text{dist}(X, \ell)} = e < 1,$$

where F is a focus, ℓ the corresponding directrix, and e the eccentricity. If both ellipses share the focus F with directrices ℓ_1, ℓ_2 and eccentricities e_1, e_2 , then a common point X satisfies

$$e_1 \cdot \text{dist}(X, \ell_1) = |FX| = e_2 \cdot \text{dist}(X, \ell_2).$$

The equation $e_1 \cdot \text{dist}(X, \ell_1) = e_2 \cdot \text{dist}(X, \ell_2)$ defines at most two lines (the signed-distance functions are affine, so we get $\pm$ cases). But X and F must lie on the same side of each directrix, so only *one* line is relevant. The intersection of one line with an ellipse contains at most two points. $\square$

Problem 2. (IMC 2009, Day 1, Problem 1b) Continuous $f \leq g$ on $\mathbb{Q} \Rightarrow f \leq g$ on $\mathbb{R}$.

Solution. The function $g - f$ is continuous on $\mathbb{R}$ and $(g - f)(r) \geq 0$ for all rational r . Every real x is a limit of a sequence r_n of rationals, so

$$(g - f)(x) = \lim_n (g - f)(r_n) \geq 0,$$

by continuity. $\square$

Problem 3. (IMC 2009, Day 1, Problem 3) Graph on n vertices: $\sum a_i^2 = n^2 - n$; shortest cycle length k .

Solution. *Answer:* $k = 5$ is the only possible value.

Let G be the friendship graph; the hypothesis is that the diameter is at most 2, no vertex has degree $n - 1$, and $\sum a_i^2 = n^2 - n$.

Count walks of length 2: the number of ordered triples (v_i, v_j, v_l) with $v_i v_j, v_j v_l \in E$ equals $\sum_j a_j^2 = n^2 - n$.

Construct an injection f from ordered pairs (v_i, v_j) of distinct vertices (there are $n^2 - n$ of them) to such walks: if $v_i v_j \notin E$, pick any common neighbour v_l and set $f(v_i, v_j) = (v_i, v_l, v_j)$; if $v_i v_j \in E$, set $f(v_i, v_j) = (v_i, v_j, v_i)$. Injectivity is easy: a walk (v_i, v_j, v_l) with $i \neq l$ can only come from (v_i, v_l) , and (v_i, v_j, v_i) only from (v_i, v_j) .

So $\sum a_i^2 \geq n^2 - n$, with equality iff f is a bijection, i.e. *exactly one* common neighbour for each non-edge and *no* common neighbour for each edge. Equivalently, G has no triangles (C_3) or 4-cycles (C_4).

If G has no cycle, it is a forest; but a forest of diameter ≤ 2 is a star, which contradicts the no-vertex-of-degree- $n - 1$ hypothesis. So G has cycles, all of length ≥ 5 . If a cycle had length ≥ 6 , two opposite vertices on it would be at distance ≥ 3 along the cycle, contradicting diameter ≤ 2 . Hence the shortest cycle has length exactly 5. The 5-cycle C_5 itself realizes all conditions. $\square$

Problem 4. (IMC 2009, Day 2, Problem 2) $f \in C^2[0, \infty)$, $f(0) = 1, f'(0) = 0, f'' - 5f' + 6f \geq 0 \Rightarrow f(x) \geq 3e^{2x} - 2e^{3x}$.

Solution. Factor the operator: $f'' - 5f' + 6f = (D - 3)(D - 2)f \geq 0$. Set $g(x) := f'(x) - 2f(x)$. Then

$$g'(x) - 3g(x) \geq 0, \quad (g(x)e^{-3x})' \geq 0,$$

so $g(x)e^{-3x} \geq g(0)e^0 = f'(0) - 2f(0) = -2$, i.e. $f'(x) - 2f(x) \geq -2e^{3x}$.

Apply the same trick again: $(f(x)e^{-2x})' = (f'(x) - 2f(x))e^{-2x} \geq -2e^x$, so $(f(x)e^{-2x} + 2e^x)' \geq 0$. Hence

$$f(x)e^{-2x} + 2e^x \geq f(0) + 2 = 3,$$

which rearranges to $f(x) \geq 3e^{2x} - 2e^{3x}$. $\square$

Problem 5. (IMC 2008, Day 2, Problem 1) $x^{2k} - x^k + 1 \mid x^{2n} + x^n + 1 \Rightarrow x^{2k} + x^k + 1 \mid x^{2n} + x^n + 1$.

Solution. Note $x^{2k} - x^k + 1 = \Phi_{6k}(x)/\Phi_{2k}(x)$, but a cleaner approach uses roots directly.

The complex root $\omega = e^{i\pi/(3k)}$ satisfies $\omega^{2k} - \omega^k + 1 = 0$ (since $\omega^k = e^{i\pi/3}$ and $e^{2i\pi/3} - e^{i\pi/3} + 1 = 0$). By the divisibility hypothesis, $\omega^{2n} + \omega^n + 1 = 0$, so $\omega^n = e^{\pm 2\pi i/3}$. Setting $n/k = r$, this gives $e^{i\pi r} = e^{\pm 2\pi i/3}$, i.e. $n = (6c \pm 2)k$ for some integer c .

Now check that any root of $x^{2k} + x^k + 1$ is a root of $x^{2n} + x^n + 1$. The roots of $x^{2k} + x^k + 1$ are $\zeta = e^{i(2\pi/3 + 2\pi s)/k}$. various k -th roots, all satisfying $\zeta^k = e^{\pm 2\pi i/3}$. Then $\zeta^n = \zeta^{(6c \pm 2)k} = (\zeta^k)^{6c \pm 2} = e^{\pm 2\pi i(6c \pm 2)/3}$. Since $6c \pm 2 \equiv \pm 2 \pmod{3}$, $\zeta^n = e^{\pm 4\pi i/3} = e^{\mp 2\pi i/3}$. Therefore

$$\zeta^{2n} + \zeta^n + 1 = e^{\mp 4\pi i/3} + e^{\mp 2\pi i/3} + 1 = 0. \quad \square$$

Problem 6. (IMC 2009, Day 2, Problem 3) $A^2B + BA^2 = 2ABA \Rightarrow AB - BA$ nilpotent.

Solution. Let $X = AB - BA$. Compute the commutator with A :

$$AX - XA = (A^2B - ABA) - (ABA - BA^2) = A^2B + BA^2 - 2ABA = 0.$$

So X commutes with A . Then for every $m \geq 0$,

$$X^{m+1} = X^m \cdot (AB - BA) = A(X^m B) - (X^m B)A,$$

and taking trace (which is cyclic) gives $\text{trace}(X^{m+1}) = 0$. The values $\text{trace}(X), \text{trace}(X^2), \dots, \text{trace}(X^n)$ being all zero determine the eigenvalues of X (Newton's identities), forcing every eigenvalue to be 0. Hence X is nilpotent: $X^n = 0$. $\square$

Problem 7. (IMC 2008, Day 1, Problem 6) $Q(n, d) = \#\{\sigma : D(\sigma) = d\}$ is even for $d \geq 2n$.

Solution. Consider the $n \times n$ determinant

$$\Delta(x) = \det(x^{|i-j|})_{i,j=1}^n.$$

By the Leibniz formula,

$$\Delta(x) = \sum_{\sigma \in S_n} (-1)^{\text{inv}(\sigma)} x^{D(\sigma)},$$

so the coefficient of x^d in $\Delta(x)$ has the same parity as $Q(n, d)$.

Evaluate $\Delta(x)$ by row-reduction: subtract $x \cdot (\text{row } i-1)$ from row i for $i = n, n-1, \dots, 2$, in that order. The resulting matrix is upper-triangular with diagonal entries $1, 1-x^2, 1-x^2, \dots, 1-x^2$, so

$$\Delta(x) = (1-x^2)^{n-1}.$$

This polynomial has degree $2(n-1) < 2n$, so its coefficient of x^d is 0 for all $d \geq 2n-1$. In particular it is even for $d \geq 2n$, and so is $Q(n, d)$. $\square$

Problem 8. (IMC 2021, Day 2, Problem 5) $2021B = A^m + B^2$ solvable with symmetric B for every $m \geq 1 \Rightarrow |\det A| \leq 1$.

Solution. Take $m = 1$: $A = 2021B_1 - B_1^2$ is symmetric (since B_1 is), so A is diagonalizable with real eigenvalues.

Now fix $m \geq 1$ and let λ be a real eigenvalue of A . In a common eigenbasis (using diagonalizability of both A and B_m , and that B_m^2 commutes with A^m via the relation), there is a real eigenvalue μ of B_m such that

$$2021\mu = \lambda^m + \mu^2, \quad \text{i.e.} \quad \mu^2 - 2021\mu + \lambda^m = 0.$$

The discriminant must be ≥ 0 :

$$2021^2 - 4\lambda^m \geq 0, \quad \lambda^m \leq \frac{2021^2}{4}.$$

If $|\lambda| > 1$, taking m even and large gives a contradiction. So $|\lambda| \leq 1$ for every eigenvalue, and $|\det A| = \prod |\lambda_i| \leq 1$. $\square$

Problem 9. (IMC 2009, Day 1, Problem 4) Convex c_k , $q(z) = \sum c_k a_k z^k$, $p(z) = \sum a_k z^k$. Show $\max_{|z| \leq 1} |q| \leq \max_{|z| \leq 1} |p|$.

Solution. By the maximum principle reduce to $|z| = 1$. For $|z| = 1$ (so $1/\bar{z} = z$) the Cauchy formula gives $a_j = \frac{1}{2\pi} \int_S \frac{p(z)}{z^j} |dz|$ for $0 \leq j \leq n$ and 0 otherwise, where $S = \{|z| = 1\}$. Thus

$$2\pi q(w) = \int_S K(w/z) p(z) |dz|, \quad K(u) := \sum_{j=-n}^n c_{|j|} u^j = c_0 + 2 \sum_{j=1}^n c_j \text{Re}(u^j).$$

(We may add the conjugate terms for $j < 0$ since the corresponding integrals vanish.)

Claim: $K(u) \geq 0$ for $u \in S$.

Set $d_k := c_{k-1} - 2c_k + c_{k+1} \geq 0$ (with $c_{n+1} = 0$), so the convexity hypothesis says $d_k \geq 0$. One checks by induction (the second difference is the Abel summation) that

$$K(u) = \sum_{k=1}^n d_k F_k(u), \quad F_k(u) = \sum_{j=-(k-1)}^{k-1} (k-|j|) u^j.$$

The function F_k is the *Fejér kernel*: for $|u| = 1$,

$$F_k(u) = \left| 1 + u + u^2 + \cdots + u^{k-1} \right|^2 \geq 0.$$

Hence $K \geq 0$ on S , and $\int_S K(u) |du| = 2\pi c_0 = 2\pi$. Therefore

$$2\pi |q(w)| \leq \int_S |K(w/z)| \cdot |p(z)| |dz| \leq M_p \int_S K |du| = 2\pi M_p,$$

where $M_p = \max_{|z|=1} |p(z)|$. □

Problem 10. (IMC 2021, Day 2, Problem 6) No injective homomorphism $\varphi : \text{GL}_2(\mathbb{Z}/p\mathbb{Z}) \rightarrow S_p$.

Solution. For $p = 2$: $|\text{GL}_2(\mathbb{Z}/2\mathbb{Z})| = 6 > 2 = |S_2|$, so no injection exists by cardinality.

For odd prime p , consider the matrices

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \text{GL}_2(\mathbb{Z}/p\mathbb{Z}), \quad B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \in \text{GL}_2(\mathbb{Z}/p\mathbb{Z}).$$

Direct computation: $A^p = \begin{pmatrix} 1 & p \\ 0 & 1 \end{pmatrix} = I$ and $A^k \neq I$ for $1 \leq k < p$, so A has order p . The matrix B is central with order 2. They commute, and AB has order $\text{lcm}(p, 2) = 2p$.

If φ is an injective homomorphism into S_p , then $\varphi(AB)$ has order $2p$ in S_p . The order of a permutation is the lcm of its cycle lengths; cycle lengths are $\leq p$ in S_p . For order $2p$ the permutation needs cycles whose lcm is $2p$, in particular a p -cycle and a 2-cycle, requiring $p + 2 \leq p$ elements — impossible. (More precisely: the only permutations in S_p of order divisible by p are p -cycles, of order exactly p .) Contradiction. □